Patient-adaptive conformal prediction sets for reliable single-channel EEG sleep staging

Zhiwen Song[a], Yuhao Peng[a,*]

[a]*School of Mathematics and Physics, China University of Geosciences (Wuhan), 388 Lumo Road, Hongshan District, Wuhan, 430074, China*

Abstract

Automatic sleep staging from a single EEG channel serves as a first pass in clinical review, but the predicted stage alone does not say which epochs need a technologist's attention. Conformal prediction turns a stager into a set-valued predictor with a distribution-free guarantee that the true stage lies in the returned set, which is what that need requires. The guarantee is marginal over epochs, however, and patients differ: a model can hold 90% pooled coverage while leaving nearly half of its patients below nominal, and the epochs where it fails are those review would otherwise catch. We study this gap for five-class staging on Sleep-EDF, using subject-wise folds that test every subject once, and report coverage per patient, per stage and over time. Three results follow. First, scoring a forward–backward posterior over each recording instead of the raw softmax is the largest efficiency gain available: the resulting deficit score cuts the mean set from 2.54 to 1.53 stages per epoch at comparable coverage and raises adjacent-epoch consistency from 0.68 to 0.94, though it under-covers N1 unless thresholds are conditioned on stage. Second, one adaptive conformal inference stream per patient, with a step size chosen by leave-one-subject-out simulation on calibration nights, raises worst-patient coverage from 0.774 to 0.886 without tuning on test labels. Third, sequence-level set construction, derived and proved here, does not improve on the per-epoch score. The sets convert into a direct estimate of review workload, from 76% to 16% of epochs; code and protocol are released.

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*Corresponding author.

Email address: pngyuo@cug.edu.cn (Yuhao Peng)

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1. Introduction

Sleep staging is the foundation of every clinical sleep study. A technologist scores each 30 s epoch of polysomnography into one of five stages — wake (W), N1, N2, N3 and REM — and the resulting hypnogram drives the diagnosis of insomnia, narcolepsy, sleep apnoea and parasomnias, as well as the assessment of treatment response. Manual scoring is slow and only moderately reliable, with inter-scorer agreement on N1 and on transition epochs notoriously poor [1]. Automatic staging has therefore become a standard first pass, and recent single-channel models reach epoch accuracy in the mid-80s on public cohorts [2, 3, 4, 5], with rhythm, multi-scale and contrastive designs continuing to raise it [6, 7].

The reviewing technologist is the part of that workflow a point prediction does not serve. A stager that is right nine epochs in ten still returns a single label for every epoch, including the ones it should not be trusted on, so the reviewer either re-scores the whole night or trusts a number that was never a statement about reliability. What the review step needs is a per-epoch account of which stages remain plausible after the signal has been read, so that attention can be spent where the model is genuinely undecided. This is a signal-analysis question before it is a machine-learning one: the decision has to be made from a short single-channel epoch, it has to be consistent with the temporal structure of the hypnogram, and it has to be usable on a patient the model has never seen.

Cohort averages hide two properties that matter at the point of review.

The first is *subject shift*. A model is trained on a cohort and then applied to a patient it has never seen. Sleep architecture varies enormously between individuals — total sleep time, N3 fraction, REM latency and the amplitude and spectral content of the EEG all change with age, medication and comorbidity — and a model that is 85% accurate on the cohort can be markedly worse on a particular patient. A cohort mean does not tell a clinician which patients those are. Per-epoch confidence scores (softmax confidence, Monte-Carlo dropout, deep ensembles, evidential heads) produce a number per epoch but no statement about how often the true stage is actually inside whatever set of stages the model is willing to commit to.

The second is *temporal dependence*. Sleep stages persist for minutes. Scored epoch by epoch, a model's errors cluster, and any per-epoch un-

certainty estimate inherits that clustering: it is confident through a stable stretch of N2 and uncertain precisely at the transitions that clinicians care about. Per-epoch prediction sets can also flicker between adjacent epochs even when the underlying signal has not changed, which is clinically incoherent, because the hypnogram is a sequence rather than a bag of epochs.

Conformal prediction (CP) [8, 9] is the natural framework for the first problem. Given a calibration set, split CP turns any model into a set-valued predictor with a finite-sample, distribution-free guarantee that the true label lies in the set with probability at least $1 - \alpha$. In sleep staging this is exactly the statement a clinician needs: "with 90% confidence, this epoch is one of these two stages." CP has also been studied as a target for network design and compression rather than only as a wrapper around a fixed model [10]; we deliberately keep the backbone frozen, so that every statement below is about the stager as it would be deployed and not about a network retrained to make the sets smaller. Recent work has begun to bring CP to EEG and to sleep, and the closest results are sobering: split CP on EEG event classification loses a large fraction of its coverage under patient shift, and per-patient recalibration recovers most of it [11]. Selective prediction for sleep staging [12] takes a related but different route — it decides which epochs to accept, subject to an empirical risk target, without producing a coverage-guaranteed set.

What is missing is (i) an evaluation of CP for sleep staging that reports the distribution of coverage across patients rather than its mean, and (ii) a procedure that delivers the per-patient statement a clinician needs while respecting the information a deployment actually has: a frozen model, a handful of scored nights from the cohort it was trained on, and labels that arrive after each prediction because a technologist reviews the hypnogram anyway.

This paper provides both, and is deliberately explicit about what the data can and cannot support. We build on Sleep-EDF [13, 14] with subject-wise folds that partition the cohort, so that every one of the 20 subjects is tested exactly once and no test epoch shares a subject with any calibration or training epoch. We evaluate two backbones, several nonconformity scores, and three families of adaptation, and we separate the claims that carry a guarantee from those that are merely measured.

The gap, and four findings.

1. **The marginal guarantee is not the clinical one.** At $\alpha = 0.1$,

pooled split CP reaches 0.898 coverage, close to nominal, while 45% of test subjects fall below the nominal level and the worst subject is covered at 0.774. An epoch-level random split — the protocol common in the machine-learning literature — hides this entirely: calibration and test epochs then come from the same subjects, so a coverage figure computed per subject reflects the shared epochs rather than the subject and is not a deployment statistic.

2. **Score smoothing is what buys efficiency.** Running forward–backward recursions over the HMM implied by the training labels and scoring the *posterior deficit* $\max_k \gamma_t(k) - \gamma_t(Y_t)$ cuts the set size by about forty per cent relative to APS on the raw softmax at comparable coverage (1.53 versus 2.54 labels per epoch, at 0.889 versus 0.898), and raises temporal consistency from 0.68 to 0.94. Because a singleton set is an epoch the reviewer can accept, set size is also the review workload: the fraction of epochs whose set contains more than one stage falls from 0.76 to 0.16. The gain is not uniform across stages, and Section 5.2 reports both the trade-off and the per-stage threshold that removes it at a modest cost in set size. The smoothing costs $O(LK^2)$ per recording and uses no labels at test time.
3. **One adaptation stream per patient shrinks the spread.** Running adaptive conformal inference (ACI) with an independent stream per patient, and choosing the step size by leave-one-subject-out simulation on the calibration subjects, raises worst-patient coverage from 0.774 to 0.886 without tuning on the test patients. The mechanism is visible in the per-patient coverage vectors: easy patients move down toward nominal, hard patients move up.
4. **Sequence-level set construction does not help here.** We derive two dynamic-programming constructors, one centred on the minimum-cost path and one projecting an exact block-level conformal set. We prove that the first reduces exactly to per-epoch CP on the deficit score when the transition weight is zero, and that its position-centred variant cannot return larger sets; at positive weight the measured sets do grow, and we report that as a measured effect rather than a proved one. We prove that the second is simultaneous by construction, and show empirically that it is vacuous on sleep blocks. We report these as negative results with proofs, because a reader who knows the literature will otherwise assume the constructions were not tried.

Relation to prior work. The procedure is assembled from parts that exist separately, and we say so plainly rather than claim each part. Gibbs and Candès [15] introduced ACI and proved long-run coverage for a single stream under arbitrary shift; running their recursion once per patient is what turns that pooled statement into a per-patient one, and Corollary 1 records the inheritance under their step-size conditions. Angelopoulos et al. [16] analyse decaying schedules; the default schedule here fails their sufficient conditions, which Section 5.4 states and quantifies rather than papers over. Barber et al. [17] and Tibshirani et al. [18] extend CP beyond exchangeability, but the resulting bounds are expressed in terms of the size of the shift or of a known weighting function, neither of which is available at deployment, and neither gives a per-patient statement. Chatterjee et al. [11] is the closest empirical predecessor: they measure the collapse of EEG coverage under patient shift and repair it with per-patient recalibration, which needs labelled epochs from the test patient before deployment. The warm-up variant studied in Section 3.8 is exactly their requirement, so we evaluate it rather than assume it; the ACI variant replaces the labelled warm-up with labels that arrive one at a time, which is what a review workflow already provides. On the sleep side the models we use are standard and the contribution is not architectural. What we contribute is the signal-processing layer that turns a per-epoch posterior into a clinically readable object: a forward–backward posterior computed from the epoch probabilities and the sleep-stage transition structure, a nonconformity score that is monotone in the posterior deficit and therefore inherits both, a calibration rule that respects the patient as the unit of deployment, and an evaluation that reports the quantities a review workflow actually consumes — coverage per patient, set size as review workload, and temporal consistency of the sets along the hypnogram.

The remainder of the paper reviews related work (Section 2), fixes notation, the CP background and the method (Section 3), describes the experimental setup (Section 4), reports results (Section 5) and discusses implications, limitations and conclusions (Section 6).

2. Related work

2.1. Automatic sleep staging

Single-channel EEG sleep staging is dominated by compact convolutional and recurrent architectures. Lawhern et al. [2] introduced EEGNet, a depthwise-separable convolutional network that remains a standard baseline;

Supratak et al. [19] and Phan et al. [3] model the sequence of epochs with recurrent layers over per-epoch convolutional features; Perslev et al. [4] use a fully convolutional network over the whole night; and Phan et al. [5] replace recurrence with self-attention and add an entropy-based uncertainty estimate. Recent work in this journal continues the same line on the same corpus, with rhythm-aware contrastive representation learning [6] and adaptive multi-scale convolution with local attention [7], and statistical modelling of the sleep EEG itself [20]. End-to-end networks can learn directly from raw single-channel EEG [21], and hybrid convolutional–HMM designs recover temporal continuity for point prediction [22]; neither construction supplies a calibrated set whose coverage is evaluated per patient. A broad evaluation of contemporary staging methods also illustrates how strongly reported performance depends on the dataset, preprocessing, and evaluation protocol [23]. Reported accuracy on Sleep-EDF is typically 82 % to 87 % with Cohen's kappa around 0.75–0.82; N1 remains the hardest class, with F1 often below 0.5, reflecting the poor inter-scorer agreement of that stage [1].

Evaluation in this literature is usually epoch-level accuracy under a subject-wise split, sometimes with a separate held-out cohort. Uncertainty, when reported, is reported as a calibration error [24, 5] rather than as a coverage guarantee, and almost always in aggregate over epochs.

2.2. Uncertainty quantification for sleep staging

Monte-Carlo dropout, deep ensembles, evidential deep learning and Bayesian last layers all produce a per-epoch confidence, and Phan et al. [5] use predictive entropy to flag epochs for review. These scores rank epochs by difficulty but carry no statement that the true stage is inside the returned set with a given probability. Selective prediction for sleep staging has been studied by Song et al. [12], who learn an error-ranking score and accept a fixed fraction of epochs subject to an empirical risk target, deferring the remainder to a larger model; this is complementary to our work, since selective prediction returns a point label plus an abstention whereas conformal prediction returns a set with a distribution-free coverage guarantee, but the risk control is empirical rather than finite-sample. Conformal prediction has also been embedded as a component of a multiple-instance sleep pipeline [25], without a study of coverage under subject shift. Uncertainty-guided pipelines such as U-PASS use uncertainty operationally during sleep-model development [26]; the present work instead evaluates the realised coverage and workload of the deployed set predictor. Recent work on sleep staging has

emphasised the distinction between aleatoric uncertainty, reflecting genuine ambiguity among scorers or signals, and epistemic uncertainty, reflecting limited model knowledge [27]. Conformal sets address the second object directly by controlling the error rate of a predictive set, while their purity and workload reveal the first.

The evaluation gap. Expected calibration error, maximum calibration error and the Brier score measure the agreement between confidence and accuracy in aggregate. They say nothing about whether a *set* contains the truth, and they are computed on the same pooled sample that hides subject heterogeneity. In our experiments the EEGNet backbone, with an ECE of 0.030, still leaves its worst patient 12.8 points below nominal coverage, which is the practical failure mode that calibration metrics do not detect.

2.3. Conformal prediction

Split conformal prediction [8, 28, 9] gives finite-sample marginal coverage under exchangeability. Adaptive prediction sets (APS) and their regularised variant (RAPS) [29, 30] adapt set size to the difficulty of each example; Mondrian and class-conditional CP [31] give per-class coverage, which on our cohort shrinks sets rather than enlarging them but leaves a residual patient gap because patients differ in ways that are not their stage mix; weighted CP [18] handles a known covariate shift. None of these provides a per-patient guarantee when the exchangeable unit is the patient rather than the epoch, and Barber et al. [17] show that without exchangeability only a weaker, non-vanishing bound is available.

Adaptive conformal inference [15] replaces exchangeability with a long-run coverage guarantee along a stream, updating the target miscoverage after each observed label; Angelopoulos et al. [16] analyse the step-size schedule and show that decaying steps with $\sum_t \gamma_t = \infty$ and $\sum_t \gamma_t^2 < \infty$ give the strongest guarantee. Standard ACI runs a single stream over all observations, so its guarantee is again pooled; we instead run one stream per patient, which is the natural unit of clinical deployment and which, to our knowledge, has not been evaluated for five-class sleep staging.

Structured and non-exchangeable data. Nettasinghe et al. [32] extend CP to hidden Markov models by partitioning the process into exchangeable blocks via de Finetti's theorem for Markov chains, producing a confidence set for a sequence through the filtering recursion. Our block-level construction shares

the motivation — blocks, not epochs, are the exchangeable unit — but differs in what is produced: they construct a set of sequences, whereas we project a scalar block-level conformal set onto each position with a forward–backward dynamic program, which gives simultaneous per-position coverage at $O(LK^2)$. We show in Section 5.5 that this projection, while valid, is too loose to be useful, which is itself informative for anyone considering the construction.

Time-series conformal prediction. A parallel literature targets real-valued forecasting under temporal dependence. EnbPI [33] bootstraps the residuals of a sequential predictor and calibrates a quantile of their running maximum, so the interval adapts to serial correlation without exchangeability; CopulaCPTS [34] models the joint distribution of multi-step residuals with a copula so that coverage holds for the whole forecast path; and Zaffran et al. [35] update the threshold online for non-stationary series. Our target is a categorical sequence with strong regime structure, and the temporal information enters through the HMM posterior rather than through a residual process. The path-level coverage those methods target is what our SeqCP-Block constructor formalises, and Section 5.5 reports that on sleep blocks it is vacuous.

Healthcare and EEG. Chatterjee et al. [11] show that split CP on EEG event classification loses over 20 coverage points under patient shift and that per-patient recalibration recovers most of it; this is the closest prior result to our adaptive component, and we treat it as such. Our contribution there is not the observation that personalisation helps but the deployment protocol: how much to personalise, chosen by leave-one-subject-out simulation on the calibration cohort rather than on the test patients; a streaming variant that needs only one label at a time; and an evaluation that reads coverage against set size for a five-class, strongly imbalanced task under subject-wise folds.

3. Materials and methods

Figure 1 summarises the pipeline. A frozen backbone produces a per-epoch distribution; an HMM fitted to the training labels smooths it over the recording; a nonconformity score turns the smoothed posterior into a scalar level; and one of three calibration rules — pooled split CP, a subject-adaptive warm-up, or a per-patient adaptive stream — turns that level into a prediction set. We describe each step and state the assumption it needs.

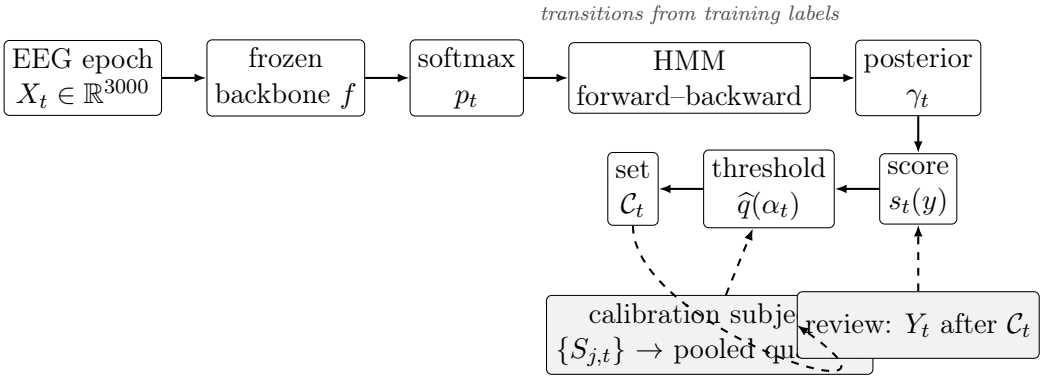


Figure 1: Overview. The backbone is frozen and the HMM transitions are estimated from training labels. The threshold comes from the calibration subjects; the online rule uses the reviewed label as feedback. The dashed path exists only in the per-patient adaptive regime.

3.1. Data

Sleep-EDF. We use the Sleep-EDF Expanded corpus from PhysioNet [13, 14], single-channel Fpz–Cz EEG at 100 Hz. Two cohorts are used. *Sleep-EDF-20* is the 20-subject cohort (SC400–SC419) commonly used in the sleep-staging literature: 39 recordings, 42 368 epochs after cropping, of which 42 307 carry a stage label (the remaining epochs sit inside the cropped window but fall between scored annotations). The *extended cohort* adds the remaining cassette recordings (SC400–SC482), giving 78 subjects and 153 recordings, and is used for the statistical-power analysis in Section 5.7.

Telemetry cohort. Sleep-EDF also contains a telemetry subset (ST700–ST721, 22 subjects, 44 recordings) recorded with different hardware on a different patient group. It is never used for training or calibration in the cross-cohort experiment, so it provides a genuine hardware-and-population shift rather than a resampling of the same distribution.

Preprocessing. Each recording is band-pass filtered to 0.5 Hz to 30 Hz, segmented into 30 s epochs, and z-scored per recording. A label is assigned to an epoch when at least 50% of it overlaps a scored annotation. R&K stage 4 is merged into N3 following the AASM rule [36], because keeping it as a separate class would create a sixth category with a handful of epochs and no modern counterpart.

Each recording is cropped to the scored sleep period plus 30 minutes of wake on either side. Keeping every annotated epoch instead yields a cohort of over 112 000 epochs that is 65% wake, which matches no published result and distorts the class priors; the crop reproduces the canonical Sleep-EDF-20 cohort with priors W 19.6%, N1 6.6%, N2 42.1%, N3 13.5% and REM 18.2%. The crop is applied identically to every cohort and is part of the released code. Subject-wise folds, training, and the evaluation metrics are described in Section 4.

3.2. Notation and problem setting

Let i index subjects, r recordings and t epochs. A subject is one patient (one or two recordings), and we use the two words interchangeably. Each epoch has an input $X_t \in \mathbb{R}^{3000}$ (30 s of a single EEG channel at 100 Hz) and a label $Y_t \in \mathcal{Y} = \{W, N1, N2, N3, REM\}$ with $|\mathcal{Y}| = 5$. A backbone f, trained on subjects disjoint from the test subject, outputs $p_t = \mathrm{softmax}(f(X_t)) \in \Delta^4$, where Δ^4 is the probability simplex. We are given labelled calibration data from subjects disjoint from the test subject, and we must output a set $\mathcal{C}_t \subseteq \mathcal{Y}$ such that

$$\mathbb{P}\big(Y_t \in \mathcal{C}_t\big) \;\geq\; 1 - \alpha \tag{1}$$

at a nominal level α, while keeping $|\mathcal{C}_t|$ small, because every epoch with $|\mathcal{C}_t| > 1$ is an epoch a human has to look at.

Two remarks fix what Equation (1) does and does not say. First, the probability is over the draw of calibration and test data, not over subjects: it is a *marginal* statement, and a predictor can satisfy it while failing on a subset of patients. Second, in sleep staging the natural exchangeable unit is arguably the patient rather than the epoch, and the two differ exactly in the regime of interest, where patients are heterogeneous.

We report, alongside coverage, the *set size* $|\mathcal{C}_t|$, the *singleton rate* $\mathbb{P}(|\mathcal{C}_t| = 1)$, the *review workload* $\mathbb{P}(|\mathcal{C}_t| > 1)$, the fraction of epochs with more than one admissible stage, the *empty rate* $\mathbb{P}(|\mathcal{C}_t| = 0)$, which also returns an epoch to the reviewer and which the workload does not count, and a *temporal consistency* statistic, the mean Jaccard overlap of the sets at adjacent epochs,

$$\mathrm{TC} = \frac{1}{|\mathcal{P}|} \sum_{(t,t+1)\in\mathcal{P}} \frac{|\mathcal{C}_t \cap \mathcal{C}_{t+1}|}{|\mathcal{C}_t \cup \mathcal{C}_{t+1}|}, \tag{2}$$

where $\mathcal{P}$ contains adjacent pairs inside a recording, so that recording bound-

aries are never counted. TC = 1 when neighbouring sets are identical and 0 when they are disjoint.

3.3. Split conformal prediction

Given a nonconformity score $s(x, y)$ for which smaller values indicate better agreement between input x and candidate label y, split CP computes $S_j = s(X_j, Y_j)$ on n calibration epochs and takes the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile,

$$\widehat{q} = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n} \big(S_1, \ldots, S_n \big), \qquad \mathcal{C}(x) = \{ y \in \mathcal{Y} : \; s(x, y) \leq \widehat{q} \}. \tag{3}$$

If the calibration and test epochs are exchangeable, then $\mathbb{P}(Y \in \mathcal{C}(X)) \geq 1-\alpha$ and $\mathbb{P}(Y \in \mathcal{C}(X)) \leq 1-\alpha+1/(n+1)$: the guarantee is finite-sample and distribution-free, and both bounds are tight for the class of all distributions [8, 28].

3.4. Nonconformity scores

The *adaptive prediction set* score [29] is

$$s_{\text{APS}}(x, y) = \sum_{y' \in \mathcal{Y}} p_{y'}(x) \, \mathbf{1}\{ p_{y'}(x) \geq p_y(x) \}, \tag{4}$$

the cumulative probability mass of all classes at least as likely as y, so that $s_{\text{APS}} \in [0, 1]$ and smaller is better. Its set is the smallest collection of most-likely classes whose total mass reaches $\widehat{q}$. RAPS [30] adds a rank penalty $\lambda_{\text{reg}} \max(0, \text{rank}(y) - k_{\text{reg}})$ to discourage a long tail of unlikely labels.

APS is heavy-tailed near one whenever the model is badly wrong, which forces the calibration quantile up and inflates every set. For the temporal component we therefore use the *posterior deficit*

$$s_{\text{def}}(x, y) = \max_{k \in \mathcal{Y}} \gamma_x(k) - \gamma_x(y), \tag{5}$$

where γ_x is a smoothed posterior (Section 3.6). This score is bounded in $[0, 1]$, is zero exactly at the maximum-posterior stage, and measures how far a candidate is from the posterior's mode.

A *class-conditional* (Mondrian) construction [31] replaces the pooled quantile by one quantile per stage,

$$\widehat{q}_y = \text{Quantile}_{\lceil (n_y+1)(1-\alpha) \rceil / n_y} \big(\{ S_j : \; Y_j = y \} \big), \tag{6}$$

where n_y is the number of calibration epochs of stage y, and sets $\mathcal{C}(x) = \{y \in \mathcal{Y} : s(x, y) \le \widehat{q}_y\}$. The guarantee then holds within each stage, $\mathbb{P}(Y \in \mathcal{C}(X) \mid Y = y) \ge 1 - \alpha$, at the price of a smaller calibration set per class; it is the baseline we use whenever a comparison needs stage-level coverage rather than set size.

3.5. Adaptive conformal inference

When exchangeability fails, adaptive conformal inference [15] updates the target miscoverage after each observed label,

$$\alpha_{t+1} = \alpha_t + \gamma_t\big(\alpha - \mathbf{1}\{Y_t \notin \mathcal{C}_t\}\big), \qquad \mathcal{C}_t = \{y : \ s(X_t, y) \le \widehat{q}(\alpha_t)\}, \tag{7}$$

where $\widehat{q}(\alpha_t)$ is the calibration quantile at level $1 - \alpha_t$. Along an arbitrary stream, a bounded step keeps the long-run miscoverage within a term governed by the step size rather than letting it drift with the shift; Angelopoulos et al. [16] show that a decaying schedule with $\sum_t \gamma_t = \infty$ and $\sum_t \gamma_t^2 < \infty$ makes that term vanish, which is the strongest form of the guarantee and the one the corollary below assumes.

3.6. Temporal smoothing

Sleep stages are strongly autocorrelated: an epoch of N3 is far more likely to be followed by N3 than by W, and a per-epoch softmax discards that information. We fit a first-order hidden Markov model to the training labels, with transition matrix A (Laplace-smoothed) and initial distribution π, and run the forward–backward recursions in log space *within each recording* to obtain

$$\begin{aligned} \gamma_t(k) &= \frac{a_t(k) b_t(k)}{\sum_{k' \in \mathcal{Y}} a_t(k') b_t(k')}, \\ a_1(k) &= \pi_k \, p_1(k), \qquad a_t(k) = p_t(k) \sum_j a_{t-1}(j) A_{jk} \quad (t \ge 2), \end{aligned} \tag{8}$$

and $b_T(k) = 1$, $b_t(k) = \sum_j A_{kj} \, p_{t+1}(j) \, b_{t+1}(j)$. (The forward and backward messages are written a and b so that α and β keep their meanings elsewhere in the paper: the nominal level and its running ACI target, and the block transition weight.) The recursion is $O(LK^2)$ per recording.

The smoothing uses no test labels and no test-time adaptation: it is a deterministic transformation of the model's own output. If calibration and

test epochs are exchangeable under the score $s(\cdot,\cdot)$, they remain exchangeable under $s(\gamma(\cdot),\cdot)$, so split CP stays valid and the change is one of *efficiency*, not of validity. That argument is the reason we prefer smoothing to a post-hoc repair of the sets, which would break the exchangeability argument it relies on.

3.7. Pooled calibration and its failure mode

Pooled calibration guarantees that $1-\alpha$ of *all* epochs are covered, not that $1-\alpha$ of a given patient's epochs are. With C the set of calibration subjects and $S_{j,t}$ the score of epoch t of subject j,

$$\widehat{q}_g = \text{Quantile}_{\lceil (n_C+1)(1-\alpha)\rceil / n_C}\left(\{S_{j,t}\}_{j\in C}\right), \tag{9}$$

which is Equation (3) applied to the pooled scores. Section 5.1 quantifies how far per-patient coverage departs from nominal under this rule.

3.8. Subject-adaptive calibration with a warm-up

If a new subject supplies m labelled epochs before deployment, those epochs define a local threshold $\widehat{q}_i$ by the same rule, and we deploy

$$\widehat{q}_i(\lambda) = (1-\lambda)\widehat{q}_g + \lambda\widehat{q}_i, \qquad \lambda \in [0,1]. \tag{10}$$

The shrinkage matters because a warm-up is short: with $m \approx 1/\alpha$ the local threshold is an extreme order statistic and its sampling variance is large, so $\lambda = 1$ trades a bias reduction for a variance increase.

Proposition 1 (Shrinkage trade-off)**.** *Let subject i have score CDF F_i that is ℓ-Lipschitz in a neighbourhood of its $(1-\alpha)$-quantile $q_i^\star$, let $\Delta = q_i^\star - q_g^\star$ be the subject's offset from the cohort quantile, and suppose $\widehat{q}_i$ and $\widehat{q}_g$ are approximately unbiased with variances σ^2/m and σ^2/n_C. Then for a threshold $\widehat{q}_i(\lambda)$ as in Equation* (10)*, the squared coverage gap satisfies*

$$\mathbb{E}\Big[\big(F_i(\widehat{q}_i(\lambda)) - (1-\alpha)\big)^2\Big] \;\le\; \ell^2\Big[(1-\lambda)^2\Delta^2 + \lambda^2\frac{\sigma^2}{m} + (1-\lambda)^2\frac{\sigma^2}{n_C}\Big], \tag{11}$$

whose minimiser over λ is

$$\lambda^\star = \frac{\Delta^2 + \sigma^2/n_C}{\Delta^2 + \sigma^2/n_C + \sigma^2/m}. \tag{12}$$

Proof sketch. Write $F_i(\widehat{q}_i(\lambda)) - (1-\alpha) = F_i(\widehat{q}_i(\lambda)) - F_i(q_i^\star)$, which is bounded in absolute value by $L|\widehat{q}_i(\lambda) - q_i^\star|$ by the Lipschitz condition. Expanding $\widehat{q}_i(\lambda) - q_i^\star = (1-\lambda)(\widehat{q}_g - q_g^\star - \Delta) + \lambda(\widehat{q}_i - q_i^\star)$ and taking expectations of squares under independence of the two estimates gives Equation (11); differentiating with respect to λ and setting the derivative to zero gives Equation (12). □

Proposition 1 is not a plug-in estimator: L, σ and Δ are unknown at deployment, and the Gaussian approximation to quantile estimates is asymptotic in m. We use it for what it does predict — that $\lambda^\star \to 0$ as $m \to 0$, that $\lambda^\star \to 1$ as $m \to \infty$, and that adaptation pays only when the subject's offset dominates the calibration noise, $\Delta^2 \gg \sigma^2/n_C$ — and we select λ by leave-one-subject-out simulation on the calibration subjects, scoring each candidate by the one-sided mean shortfall

$$\mathcal{L}(\lambda) = \mathbb{E}\big[\max\{0, (1-\alpha) - \widehat{\mathrm{cov}}\}\big], \tag{13}$$

so that over-coverage is not rewarded; it is paid for in set size and is not banked as a gain. Test subjects never enter this selection, which differs from the usual practice of reporting the best λ on test data.

Label-free variant. When no warm-up labels exist, each subject is described by unlabelled statistics $z_i = (\bar{p}_i,\ \overline{\max_k p_{i,k}},\ \bar{H}(p_i)) \in \mathbb{R}^7$ — the mean predicted class distribution, the mean confidence and the mean predictive entropy, with the distribution contributing five coordinates — and the k nearest calibration subjects in this space supply a local threshold, shrunk toward the global one with the same λ. This is an approximation and we report it as such: it cannot be valid in general, only when the descriptor correlates with the subject's score distribution.

3.9. Per-patient adaptive conformal inference

The warm-up regime presumes labelled epochs from the new patient before deployment. In a review workflow the opposite is true: labels arrive *after* each prediction, because a technologist reviews the automatic hypnogram. This is the feedback pattern ACI was designed for.

Running a single ACI stream over all epochs of all patients gives a pooled guarantee once more: a hard patient can be under-covered while an easy one is over-covered and the stream still converges. We therefore run one

independent stream per patient p,

$$\alpha_{p,t+1} = \alpha_{p,t} + \gamma_{p,t}\big(\alpha - \mathbf{1}\{Y_{p,t} \notin \mathcal{C}_{p,t}\}\big), \qquad \mathcal{C}_{p,t} = \{y : \; s_{p,t}(y) \le \widehat{q}(\alpha_{p,t})\}, \tag{14}$$

each starting at $\alpha_{p,1} = \alpha$ and each reading its threshold from the same pooled calibration scores, so that no patient-specific calibration data is required. The step schedule is

$$\gamma_{p,t} = \gamma_0/\sqrt{t+2}, \tag{15}$$

which decays fast enough to stop a single misclassified stretch from permanently loosening the threshold, and slowly enough to keep adapting over a night. The schedule has $\sum_t \gamma_t = \infty$ but also $\sum_t \gamma_t^2 = \infty$, so it violates the square-summability condition under which Angelopoulos et al. [16] establish their guarantee; we make *no* long-run guarantee for it, and report the realised per-patient coverage and state in Section 5.1 whether it is above or below nominal. What the procedure buys, and what we measure, is the reallocation of coverage from easy to hard patients.

The point of running one stream per patient is that it turns ACI's pooled long-run statement into a per-patient one. We state that implication explicitly, because it is the only part of the construction that carries a guarantee.

Corollary 1 (Per-patient long-run coverage)**.** *Suppose the threshold map $\widehat{q}(\alpha)$ is the fixed quantile of the pooled calibration scores, so that Equation* (14) *defines every set, and let the step schedule satisfy $\gamma_t \to 0$, $\gamma_t \le 1$, $\sum_t \gamma_t = \infty$ and $\sum_t \gamma_t^2 < \infty$. Then for every patient p the realised miscoverage rate on that patient's stream converges to α,*

$$\frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{Y_{p,t} \notin \mathcal{C}_{p,t}\} \;\longrightarrow\; \alpha \qquad (T \to \infty), \tag{16}$$

for an arbitrary sequence of inputs and labels, with no exchangeability assumption within the patient.

Proof. The update Equation (14) of stream p depends only on that stream's own outcomes and on a threshold map fixed by the calibration set, so the stream is an instance of the single-stream recursion of Angelopoulos et al. [16]. Their long-run bound, applied stream by stream, gives Equation (16). The step Equation (15) used in the experiments does not satisfy the summability hypothesis and therefore carries no claim; Section 5.4 repeats the experiment with a schedule that does. □

The scale γ_0 is a hyperparameter. We choose it by leave-one-subject-out simulation within the calibration cohort: each calibration subject in turn forms a stream, the remaining calibration subjects supply the score pool, and the γ_0 that minimises the one-sided mean shortfall Equation (13) across those held-out streams — ties broken by smaller mean set size — is applied unchanged to every test patient. The test cohort never influences the choice.

3.10. Sequence-level constructors

A per-epoch set is free to be inconsistent with its neighbours: epoch t may admit $\{W, N1\}$ while $t+1$ admits $\{N3\}$, a transition the training labels essentially forbid. We derived two constructors that repair this by scoring labels in the context of the best *sequence* through them, and we report both with their properties because one of them is a useful ablation and the other is a negative result.

SeqCP-Margin. For a block of L consecutive epochs inside one recording, transition cost $\tau(a, b) = 1 - A_{ab}$, and weight $\beta \geq 0$, define

$$\begin{aligned} M_t(y) &= \min_{y_{1:L}:\, y_t = y} \Big[\frac{1}{L} \sum_{u=1}^{L} \big(1 - \gamma_u(y_u)\big) + \frac{\beta}{L-1} \sum_{u=2}^{L} \tau(y_{u-1}, y_u) \Big], \\ M^\star &= \min_{t,y} M_t(y). \end{aligned} \tag{17}$$

Forward and backward recursions compute every $M_t(y)$ in $O(LK^2)$; a brute-force check over all K^L paths for small blocks is part of the test suite. We take $m_t = M_t(Y_t) - M^\star$ as the nonconformity of the true label, calibrate $\widehat{\delta}$ as its split-conformal quantile on calibration blocks, and predict $\mathcal{C}_t = \{y : M_t(y) - M^\star \leq \widehat{\delta}\}$. Since $Y_t \in \mathcal{C}_t \iff m_t \leq \widehat{\delta}$, this is split CP on the scalar m_t and inherits its marginal guarantee.

Proposition 2 (Reduction at $\beta = 0$)**.** *With $\beta = 0$ the dynamic program factorises: for every t and y,*

$$M_t(y) - M^\star = \frac{1}{L} \Big(s_t(y) - \min_{k \in \mathcal{Y}} s_t(k) \Big), \qquad s_t(k) = 1 - \gamma_t(k), \tag{18}$$

independently of the block. Hence SeqCP-Margin at $\beta = 0$ is exactly per-epoch split CP on the deficit score Equation (5)*, and any difference between the two at $\beta > 0$ is attributable to the transition term alone.*

Proof. At $\beta = 0$ the objective separates over positions. The minimum total cost of a path constrained to $y_t = y$ is $s_t(y) + \sum_{u \neq t} \min_k s_u(k)$, so $M_t(y) - M^\star = \frac{1}{L}(s_t(y) - \min_k s_t(k))$ after subtracting the unconstrained minimum $\frac{1}{L}\sum_u \min_k s_u(k)$. □

Proposition 3 (Monotonicity of the centring)**.** *Let $M_t^{\mathrm{pos}}(y) = M_t(y) - \min_k M_t(k)$. Then $M_t^{\mathrm{pos}}(y) \leq M_t(y) - M^\star$ for every t, y, so the position-centred sets are pointwise no larger than the globally centred ones.*

Proof. $\min_k M_t(k) \geq \min_{u,k} M_u(k) = M^\star$. □

In practice the two coincide, because the block-optimal path on a sleep block is usually constant and therefore visits every epoch, in which case $\min_k M_t(k) = M^\star$ for all t. We keep both and report that they agree rather than asserting a difference.

SeqCP-Block. Projecting a *sequence-level* conformal set onto each position gives a simultaneous guarantee with no union bound. Calibrate the block score

$$S(y_{1:L}) = \frac{1}{L}\sum_{t=1}^{L} \mathrm{APS}_t(y_t) + \frac{\beta}{L-1}\sum_{t=2}^{L} \tau(y_{t-1}, y_t) \tag{19}$$

on complete calibration blocks, obtaining $\widehat{q}$, and predict $\mathcal{C}_t = \{y : M_t(y) \leq \widehat{q}\}$ with M_t computed from the APS score matrix.

Theorem 1 (Simultaneous coverage of the projection)**.** *Let $\widehat{q}$ be the split-conformal quantile of Equation* (19) *over n_B exchangeable calibration blocks. Then with probability at least $1 - \alpha$, the true block is in the block-level set, and in that event $Y_t \in \mathcal{C}_t$ for every $t = 1, \ldots, L$.*

Proof. By Equation (3), $\mathbb{P}(S(Y_{1:L}) \leq \widehat{q}) \geq 1 - \alpha$. If the true block is in the block-level set then the path $y_{1:L} = Y_{1:L}$ is feasible, so $M_t(Y_t) \leq S(Y_{1:L}) \leq \widehat{q}$ by the definition of M_t as a minimum over paths. Hence $Y_t \in \mathcal{C}_t$ for all t simultaneously. □

Theorem 1 is the guarantee we can prove, and Section 5.5 reports what it costs. The projection of a union of sequences contains every label that *any* admissible sequence assigns to t, so it is inherently loose, and because the transition terms are non-negative penalties the block set prefers low-transition sequences. At $\beta \geq 0.5$ the penalised block set admits a constant

sequence for every label, the projection becomes the whole label set, and the guarantee is satisfied trivially. Even at $\beta = 0$ the block mean lets one well-scored neighbour absorb a badly scored epoch. We report this as a property of the construction rather than of our implementation.

3.11. What we do not claim

The nonconformity scores are calibrated, not the model: a set can cover the truth and still be clinically useless if it is always $\{N1, N2\}$, which is why set size, singleton rate and review workload are reported alongside coverage. We make no subject-conditional finite-sample guarantee under arbitrary shift, no claim that SeqCP-Block is efficient, and no clinical safety claim. The assumptions each result needs are stated where the result is used.

4. Experimental protocol

4.1. Splits

Subject-wise folds. Subjects are shuffled once and partitioned into five blocks. Fold f tests on block f, calibrates on four subjects drawn from the remaining blocks (rotated across folds), and trains on the rest. The test blocks partition the cohort, so every subject contributes exactly one test set across the five folds, and no test epoch shares a subject with any calibration or training epoch of the same fold. Per-subject statistics are computed by merging the five folds and then averaging each subject's coverage over the seeds and backbones that re-run that partition. Each subject therefore contributes exactly one coverage value to the worst-patient and dispersion statistics, and the spread we report for them is the run-to-run spread of the training, not a spread over overlapping fold summaries.

The four-subject calibration cohort is deliberate: it is the regime a hospital faces when it has a handful of scored nights, and it is where the marginal guarantee is weakest. Section 5.3 varies the size explicitly.

Random-epoch split. As a diagnostic we also train and calibrate with a 60/20/20 epoch-level random split. Calibration and test epochs then come from the same subjects, which is the protocol under which most published epoch-level numbers are computed. It is not a valid estimate of deployment performance, and we use it only to show how much of the coverage gap it hides.

Cross-cohort. For the shift experiment, models are trained and calibrated on SC subjects and tested on ST subjects, and vice versa, with no subject overlap in either direction.

4.2. Models and training

Two backbones are used throughout so that conclusions are not tied to a single architecture: `eegnet`, a depthwise-separable convolutional network in the style of Lawhern et al. [2] with 1445 parameters, and `sleepcnn`, a four-block one-dimensional VGG-style network with global average pooling and 224 069 parameters. Both take a 3000-sample epoch and output a five-way softmax.

Training uses AdamW (learning rate 10^{-3}, weight decay 10^{-4}), a cosine schedule, batch size 128, at most 40 epochs, and early stopping on a held-out subset of the training subjects. The loss is plain cross-entropy without class weights in every run reported here. Class weighting trades N1 recall against the calibration of the output distribution, which is exactly the quantity CP consumes, and mixing the two settings across tables would make coverage numbers incomparable; we therefore fix one setting and record it in each run's metadata.

In the main matrix every configuration is run with three seeds (0, 1, 2) and five folds, giving 15 runs per backbone and 30 in total; the extended cohort and the cross-cohort experiment use seed 0 only, so their numbers are single-seed estimates over five folds and two backbones, and the spread reported there is across those ten runs rather than across seeds. The run-to-run spread over seeds and backbones is reported as the uncertainty on every headline number of the main matrix, and per-subject statistics average a subject's coverage over those runs before any worst-patient or dispersion statistic is formed.

Hardware and software. All experiments run on a single NVIDIA RTX 4060 Laptop GPU with PyTorch; preprocessing, conformal procedures and dynamic programs are NumPy and run on one CPU core. Training is seeded and cuDNN is pinned to deterministic kernels, so repeating a training run reproduces its test probabilities exactly; we verified this by retraining one fold while the rest of the matrix occupied the GPU and comparing the arrays. The released code reproduces every table and figure from the raw archives with the commands recorded in the repository README.

4.3. Conformal procedures and hyperparameters

Unless stated otherwise $\alpha = 0.1$. All thresholds are computed on calibration subjects only. The procedures compared are:

- *Pooled APS* and *pooled RAPS* on the raw softmax, with the split-conformal threshold Equation (3).
- *Smoothed APS*: APS on the forward–backward posterior Equation (8).
- *Smoothed deficit*: the score Equation (5) on the same posterior.
- *Per-stage thresholds*: the Mondrian construction of Section 3.4, applied to both the raw APS score and the smoothed deficit score, so that the stage-level comparison uses the same scores as the pooled one.
- *SACP*: warm-up calibration with shrinkage Equation (10), for label budgets $m \in \{0, 10, 20, 40, 80, 160, 320\}$ and $k \in \{3, 5, 10\}$ neighbours for the label-free variant, with λ chosen by leave-one-subject-out simulation on the calibration subjects.
- *Per-patient ACI*: one stream per patient Equation (14), with γ_0 chosen by leave-one-subject-out simulation from $\{0.005, 0.01, 0.02, 0.05, 0.1\}$.
- *SeqCP-Margin* and *SeqCP-Block*: the sequence constructors of Section 3.10, for block lengths $L \in \{10, 20, 30\}$ and transition weights $\beta \in \{0, 0.5, 1, 2\}$.
- *Oracle*: per-subject thresholds computed on the test subject's own labels. Not deployable; it is the lower bound on set size at nominal coverage and shows how much room per-patient adaptation has.

HMM transitions are estimated from the training labels with Laplace smoothing. Block-level comparisons restrict all methods to exactly the epochs inside complete blocks, and the per-epoch baselines in those comparisons are recalibrated on the same calibration epochs, so the block-length comparison isolates the construction rather than the calibration sample.

Table 1: Backbone classification performance on the test subjects, over all folds and seeds (mean ± standard deviation).

Model	Accuracy	Cohen's κ	Macro-F1	ECE	Runs
`eegnet`	0.781 ± 0.042	0.698 ± 0.058	0.644 ± 0.034	0.030 ± 0.013	15
`sleepcnn`	0.803 ± 0.042	0.728 ± 0.058	0.711 ± 0.043	0.032 ± 0.017	15

4.4. Metrics

Coverage is reported marginal, per stage, and per subject; from the per-subject values we report the mean, the 10th percentile, the worst subject and the fraction of subjects below nominal. Efficiency is reported as mean set size, singleton rate and review workload, the fraction of epochs whose set contains more than one stage and would therefore need adjudication. Temporal behaviour is reported with the Jaccard consistency statistic of Section 3.2 and with sequence coverage, the fraction of complete blocks all of whose epochs are covered. Classification quality is reported with accuracy, Cohen's kappa, macro-F1 and expected calibration error (15 equal-mass bins), so that the coverage results can be read against the underlying model quality.

5. Results

All numbers are at $\alpha = 0.1$ unless stated. Coverage statistics over subjects are computed by merging per-subject coverage across the five folds and averaging it over the seeds and backbones that re-run the same partition, so each subject appears exactly once; the spread reported is the run-to-run spread of the training, not a spread over patient cohorts.

Backbone performance. Table 1 reports classification quality. Both backbones sit in the range published for Sleep-EDF on Fpz–Cz, with `sleepcnn` slightly ahead of `eegnet` on accuracy and kappa. N1 remains the weak class for both, and the per-stage breakdown in Table 3 shows that the marginal number hides class-level deviations in both directions. ECE is modest (0.030 for `eegnet`), and Section 5.1 shows that this is precisely the number that fails to detect the problem: a well-calibrated marginal predictor can still leave a patient twelve points short of nominal coverage.

5.1. The marginal guarantee is not the patient guarantee

Table 2 contrasts pooled calibration with the patient-level statistics. Pooled APS reaches 0.898 marginal coverage — nominal to within two thousandths — while the worst patient is covered at 0.774, 45% of the patients fall below $1 - \alpha$, and per-patient coverage has a standard deviation of 0.051. The quantity that matters clinically is the *patient coverage gap*,

$$\max_j \left[(1 - \alpha) - \mathrm{cov}_j\right]^+, \tag{20}$$

and it is invisible in the marginal number. The first row makes the point sharper: a random-epoch split — the protocol under which subjects are mixed between calibration and test — reports 0.901 coverage. A per-patient figure could be computed there — epochs carry their subject identifier — but it would measure the epochs shared between calibration and test rather than the subject, so the row is reported with marginal coverage and set size only. The split that makes the statistic estimable is the same split that makes the failure visible.

The class-conditional (Mondrian) row is the informative baseline, and it deserves care because it is not a straw man. Conditioning the threshold on the stage makes the mean set *smaller* — from 2.54 to 2.29 labels per epoch — because a per-class APS threshold no longer has to pay for the heavy tail of the pooled score, and it nudges the patient statistics in the right direction: worst-patient coverage rises from 0.774 to 0.787 and the per-patient standard deviation falls from 0.051 to 0.048. The nudges are small, and a paired per-patient test on the distance from nominal finds no difference from pooled calibration (smaller for 12 of 20 patients, $p = 0.43$). The mechanism is that patient difficulty is strongly associated with stage mix — every patient is hard on N1 — so a stage-conditional rule equalises patients only to the extent that they differ by stage alone. What it cannot do is adapt to a patient who is hard for reasons orthogonal to their stage mix, and on this cohort it still leaves an 0.11 gap at the worst patient and 45% of patients below nominal. The same limitation applies to the per-stage threshold on the deficit score of Section 5.2: conditioning reaches a patient only through their stage mix, so it reduces the fraction of patients below nominal from 0.60 to 0.35 but leaves the worst patient where it was (0.787 against 0.792). That residual is the target of Section 5.4.

Table 2: Coverage over patients at $\alpha = 0.1$. Per-subject coverage is merged across the five subject-wise folds, so each of the 20 subjects appears exactly once. The random-epoch row is the protocol that mixes epochs from all subjects between calibration and test; a per-patient decomposition under it would measure the shared epochs rather than the subject, so the row reports marginal coverage and set size only.

Method	Coverage	Worst subj.	p_{10} subj.	SD	Below	Set size	Singleton
Random-epoch split	0.901	–	–	–	–	2.542	–
Pooled (APS)	0.898 ± 0.038	0.774	0.822	0.0510	0.45	2.537 ± 0.207	0.167 ± 0.051
Pooled (RAPS)	0.885 ± 0.037	0.785	0.836	0.0397	0.65	1.614 ± 0.206	0.472 ± 0.178
Mondrian (per class)	0.896 ± 0.031	0.787	0.835	0.0480	0.45	2.294 ± 0.157	0.230 ± 0.062
kNN subject (APS)	0.894 ± 0.040	0.759	0.828	0.0536	0.50	2.509 ± 0.214	0.170 ± 0.049
Warm-up SACP, $m = 80$	0.897 ± 0.036	0.782	0.820	0.0496	0.40	2.552 ± 0.299	0.172 ± 0.055
Warm-up SACP, $m = 320$	0.891 ± 0.031	0.769	0.829	0.0516	0.50	2.519 ± 0.151	0.155 ± 0.045
Oracle per patient	0.901 ± 0.000	0.901	0.901	0.0004	0.00	2.492 ± 0.230	0.180 ± 0.058
Per-patient ACI (APS)	0.904 ± 0.010	0.886	0.892	0.0117	0.50	2.491 ± 0.203	0.177 ± 0.060
Per-patient ACI (deficit)	0.898 ± 0.007	0.872	0.891	0.0097	0.70	1.575 ± 0.271	0.831 ± 0.072

"Coverage" is marginal over epochs (mean ± SD over the runs); "Worst subj." is the minimum per-subject coverage over the runs; "Below" is the fraction of subjects below $1 - \alpha$; "SD" is the standard deviation of per-subject coverage. Each subject contributes one value per run, the runs being five folds × three seeds × two backbones; set size and singleton rate are averaged over the same runs.

5.2. Posterior smoothing is the largest efficiency lever

Table 8 compares nonconformity scores on the same epochs and the same calibration data. Replacing the raw-softmax APS score with the smoothed posterior deficit reduces the mean set size from 2.54 to 1.53 labels per epoch at essentially the same coverage (0.898 against 0.889, a difference within the run spread), and raises the Jaccard consistency statistic from 0.68 to 0.94. Both sit close to nominal — the deficit score 1.1 points below and APS 0.2 points below — and the residual gap of the deficit score is the finite-sample fluctuation of the split-conformal threshold — the guarantee is in expectation over calibration draws, not a statement about one calibration set — and it is smaller than the run-to-run spread. The mechanism is visible in the singleton rate, which rises from 0.17 to 0.84: most epochs are predicted as a single stage, which is the behaviour a reviewing technologist wants. The review workload — the fraction of epochs whose set contains more than one stage and would need adjudication — falls from 0.76 to 0.16 over the same runs. A second asymmetry is not in the table: pooled APS returns an *empty* set on 6.8% of epochs, where no stage clears the threshold, and those epochs also go to the reviewer even though the workload definition does not count them. The deficit score is never empty (0.000), because its score has an atom

at zero and the posterior mode always admits itself. Adding empty sets to the workload leaves the comparison unchanged (0.83 against 0.16).

The efficiency is not free at the class level. Table 3 breaks coverage down by stage. Pooled APS over-covers the rare and difficult stages — N1 at 0.96 and REM at 0.985 — because the heavy tail of its score forces a loose threshold, and it under-covers W (0.81) and N3 (0.87). Smoothing the APS score redistributes the imbalance rather than removing it: W falls to 0.75 while N1 stays at 0.95. The deficit score removes the looseness where the model is confident and pays for it on N1, whose coverage falls to 0.43: when the model is confidently wrong on an N1 epoch, the posterior deficit of the true N1 is large, so the set omits the stage exactly where the error is.

The repair is one line of calibration rather than a different score: thresholding the same deficit score per stage (Equation (6)). Every stage then lands between 0.899 and 0.909 at 1.74 labels per epoch — a quarter fewer than conditioning APS on the stage (2.29) and almost a third fewer than pooled APS (2.54). It also lowers the fraction of patients below nominal from 0.60 to 0.35, although the worst patient is unchanged (0.787 against 0.792 for the pooled deficit threshold). What conditioning costs is review workload, which rises from 0.16 to 0.49 because the sets add a second stage whenever the model's evidence is weak, and adjacent-set consistency, which falls from 0.94 to 0.88. We therefore report the unconditioned deficit score as the efficiency frontier and the per-stage version as the deployment default when an omitted stage is costly; the per-patient analysis below is marginal over stages and is unaffected.

The recursion costs nothing at inference: $O(LK^2)$ per recording, measured at 30 µs to 40 µs per epoch — a few tens of milliseconds for a whole recording — on one CPU core (`scripts/benchmark_smoothing.py` reproduces the measurement on the machine at hand).

APS on the smoothed posterior sits between the two: smoothing alone helps ($2.54 \to 2.22$ labels at 0.903 coverage, consistency 0.75), but the heavy tail of the APS score limits how much, and the deficit score removes it. RAPS narrows the tail at the cost of a rank hyperparameter that must be tuned on held-out calibration data; it reaches 1.61 labels at 0.885 coverage with 47% singletons, better than APS and short of the deficit score, and we do not pursue it further because the deficit score needs no extra hyperparameter.

The ordering does not depend on the nominal level. Recomputing both scores on the same fitted models at $\alpha \in \{0.05, 0.2\}$, the deficit score uses fewer labels at every level (2.16 against 2.97 at $\alpha = 0.05$; 1.05 against 2.00

Table 3: Coverage per stage at $\alpha = 0.1$, averaged over the 30 runs. Nominal coverage is 0.90 for every stage.

Stage	Pooled APS	Smoothed APS	Smoothed deficit	Deficit, per stage	Mondrian APS
W	0.812	0.747	0.925	0.904	0.887
N1	0.961	0.953	0.432	0.902	0.904
N2	0.899	0.924	0.933	0.909	0.897
N3	0.865	0.923	0.912	0.899	0.899
REM	0.985	0.980	0.912	0.905	0.906
Mean $\lvert\mathcal{C}_t\rvert$	2.537	2.223	1.531	1.741	2.294

Each coverage cell is the fraction of epochs of that stage whose set contains the true stage, averaged over the runs; the averages weight the stages unequally because the test blocks differ, so the run-to-run spread is in the released JSON. The bottom row is the mean set size over all epochs, the counterpart of those coverage numbers. The last two columns threshold per stage, on the deficit score and on the raw APS score respectively, so their stage coverage is near nominal by construction.

at $\alpha = 0.2$), and at $\alpha = 0.2$ it over-covers where APS under-covers (0.825 against 0.794 for a nominal 0.80). The gap in coverage between the two score families is therefore not an artefact of the one level the paper reports in its tables.

5.3. Calibration-cohort size and the label budget

Table 5 isolates the calibration-cohort axis by subsampling k of the four calibration subjects, refitting the pooled threshold, and repeating the subsample fifty times. Marginal coverage of the pooled APS threshold converges quickly in k, but the worst of its four test subjects — a per-run minimum, averaged over runs, not the merged-cohort worst patient — plateaus around 0.84, six points below nominal; the deficit score's corresponding statistic is flat in k (0.848 at $k = 1$, 0.849 at $k = 4$). The patient gap is therefore not calibration-sampling noise: it is present with a single calibration subject and essentially unchanged with all four. Adding calibration patients buys a less noisy estimate of the wrong quantity.

Figure 3 reports the label-budget regime. Proposition 1 says the optimal shrinkage weight rises with the warm-up budget m; the obstacle is estimating it. Warm-up from the new patient helps only if the shrinkage weight is chosen well, and at this cohort size leave-one-subject-out selection cannot choose it. The criterion is degenerate: it returns $\lambda = 0$ at every budget up to $m = 80$ in 16 of the 30 runs, and where it does select a positive weight the choice is

unstable across folds and seeds — at $m = 80$ the same budget selects 0, 0.25, 0.5, 0.75 and 1 in different runs. Merging per-subject coverage across folds, the curve does not improve and the largest budget degrades it: worst-patient coverage is 0.774 at $m = 0$ and 0.769 at $m = 320$, with no budget in between exceeding 0.783.

Table 4 holds the shrinkage weight fixed instead of selecting it, and shows how much of the loss is the selection rule. Short warm-ups are not marginal, they are catastrophic: at $m = 20$ the worst patient falls from 0.780 under the pooled threshold to 0.685 at $\lambda = 0.5$ and 0.590 at $\lambda = 1$. At $m = 160$ the same weights leave the worst patient at 0.788 and 0.797 against 0.772 for the pooled threshold — a gain of at most two and a half points, against the thirteen-point gap to the oracle — and at $m = 320$ the ordering reverses (0.766 pooled, 0.776 at $\lambda = 0.5$, 0.755 at $\lambda = 1$). So the degeneracy of the selection is not the whole story: even a fixed, generous weight cannot close the patient gap at this cohort size, and at small budgets it does much worse.

The mechanism is visible in what a warm-up actually contains. A warm-up is a poor sample of the night it is deployed on. At $m = 160$ it covers the first 80 minutes, in which wake accounts for 42% of epochs and REM for none, against 18% and 20% over the rest of the night (reproduced by `scripts/warmup_composition.py`), so the threshold is estimated on one stage mix and applied to another. At $m = 20$ the estimate is also built from twenty epochs, so it carries the variance term of Equation (11) on top of the bias.

We report this as a negative result rather than a design choice. With four calibration subjects there is too little held-out data to separate the gain from adapting the threshold to the new patient from the loss from estimating it on a handful of epochs, and the selected weight appears to be decided by which of the four subjects happens to be hardest. The label-free k-nearest-subject variant fails for a structural reason: with four calibration subjects, $k = 3$ already draws on most of the pool and gives the worst patient 0.759, below the pooled 0.774, while $k \geq 5$ is the whole pool and returns the pooled threshold exactly. With four subjects there is no neighbourhood to select. Section 5.4 gives the alternative that needs no warm-up labels at all.

Figure 2 puts the same sweep on the efficiency–reliability plane. Across a factor of 32 in label budget the warm-up path stays within a tenth of a label and one point of worst-patient coverage of the pooled point, and never approaches the oracle anchor, which sits at essentially the same set size but at nominal worst-patient coverage. The shortfall is therefore not a question

Table 4: Warm-up calibration at fixed shrinkage weights, $\alpha = 0.1$.

m	λ	Coverage	Worst subj.	SD	Set size
$m = 20$	0	0.899	0.780	0.050	2.541
	0.5	0.868	0.685	0.068	2.407
	1	0.882	0.590	0.104	2.763
$m = 80$	0	0.900	0.780	0.050	2.538
	0.5	0.884	0.788	0.049	2.440
	1	0.893	0.731	0.066	2.617
$m = 160$	0	0.899	0.772	0.051	2.558
	0.5	0.893	0.788	0.044	2.501
	1	0.903	0.797	0.051	2.619
$m = 320$	0	0.899	0.766	0.054	2.582
	0.5	0.892	0.776	0.051	2.513
	1	0.895	0.755	0.059	2.574

Within a block, the rows share the same evaluation epochs: the first m epochs of each patient are used only for warm-up and are excluded from every row, so the $\lambda = 0$ row is the pooled threshold applied to the post-warm-up epochs and the other rows differ from it only in the shrinkage weight. Per-subject coverage is merged across folds and averaged over seeds and backbones. Over the 30 runs, leave-one-subject-out selection returned zero for every budget up to $m = 80$ in 16 of them; the full selection histogram is in the released JSON.

of choosing a better budget: it is the distance between what a per-patient threshold could do with the patient's own labels and what any warm-up rule achieves.

5.4. Per-patient adaptation collapses the coverage spread

Table 6 is the central result. Pooled calibration leaves a spread of 0.051 across patients and a worst patient at 0.774. Running one ACI stream per patient, with the step scale chosen on held-out calibration streams, collapses the spread to 0.012 and lifts the worst patient to 0.886, at a mean set size of 2.49 against 2.54. The deficit score inherits the correction: its worst patient moves from 0.792 to 0.872 at essentially unchanged set size (1.53 to 1.58), so the efficiency gain of Section 5.2 and the spread collapse of this section compose.

The comparison is paired by patient, and we test the quantity the section is about rather than the mean. Mean per-patient coverage changes little and not significantly (0.896 to 0.904, Wilcoxon signed-rank $p = 0.84$), but the distance of a patient's coverage from nominal is smaller under ACI for 19 of

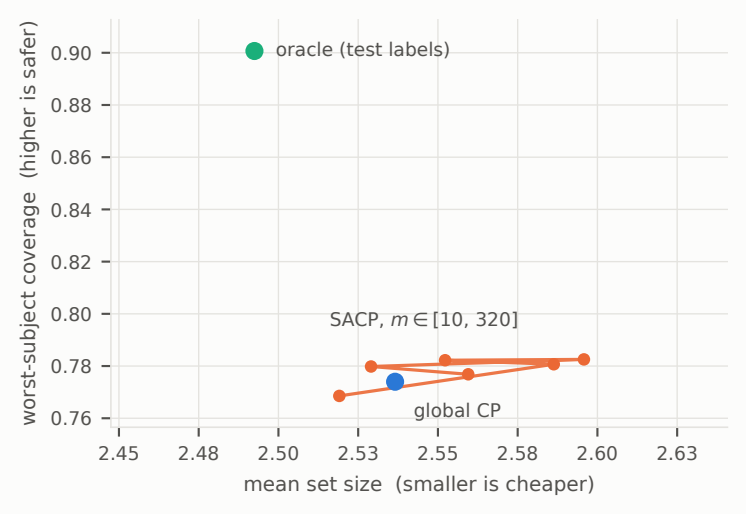


Figure 2: Worst-patient coverage against mean set size along the warm-up budget $m \in \{10, \ldots, 320\}$, with the shrinkage weight selected by leave-one-subject-out simulation at each budget. The blue point is pooled calibration, which is also the $m = 0$ rule, and the teal point is the oracle per-patient threshold computed on the test patient's own labels. The path stays next to the pooled point across the whole budget range, and the distance to the oracle is what no warm-up rule closes.

the 20 patients (median reduction 0.025, $p = 9.5 \times 10^{-6}$), and smaller than under Mondrian for 19 of 20 ($p = 3.8 \times 10^{-6}$). The one patient who moves away from nominal is above it under both rules and moves by six thousandths (from 0.908 to 0.914), well inside the run-to-run spread. Mondrian is the nearest alternative on this statistic, and it is not distinguishable from pooled calibration (12 of 20, $p = 0.43$), which is why we read the Mondrian changes in Section 5.1 as directionally right but not significant on this cohort.

Table 7 reports the same experiment at each step scale rather than at the selected one. The worst patient improves monotonically in γ_0 for both scores, from 0.810 to 0.888 (APS) and from 0.828 to 0.882 (deficit), and the mean absolute deviation from nominal falls from 0.030 to 0.006 and from 0.026 to 0.006. Larger steps are not free: the fraction of patients below nominal rises from 0.40 to 0.75 (APS), because the recursion pulls every stream toward the marginal rate, and once the spread has collapsed it is the mean, not the worst patient, that becomes the binding quantity. The grid stops at $\gamma_0 = 0.1$ for that reason — the spread reduction has saturated while the fraction below nominal is still rising. The LOSO-selected value (0.062 for APS, 0.075 for

Table 5: Coverage against the number of calibration subjects k, subsampling the available calibration subjects and repeating 50 times per run; rows are averaged over all runs, and $\pm$ is the standard deviation across runs. "worst" is the minimum per-subject coverage within a run (each run tests four subjects), averaged over runs, so it is not the merged-cohort worst patient of Table 2; it is the quantity that this sweep can hold fixed while k varies. The deficit score's worst-patient coverage is flat in k: the patient gap is not calibration-sampling noise.

	APS		Posterior deficit	
k	cov.	worst	cov.	worst
1	0.882 ± 0.037	0.819 ± 0.058	0.887 ± 0.042	0.848 ± 0.058
2	0.894 ± 0.036	0.833 ± 0.060	0.886 ± 0.043	0.847 ± 0.059
3	0.896 ± 0.038	0.837 ± 0.062	0.889 ± 0.044	0.850 ± 0.060
4	0.898 ± 0.038	0.838 ± 0.062	0.889 ± 0.044	0.849 ± 0.061

APS columns are the raw-softmax score; deficit columns use the smoothed posterior deficit. Nominal coverage is 0.900. Calibration pool: 4 subjects.

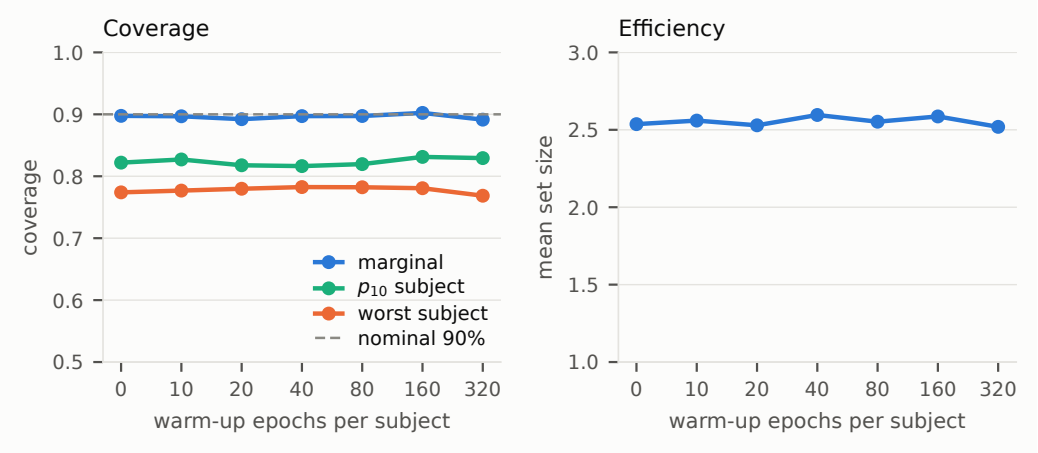


Figure 3: Left: marginal, p_{10} and worst-patient coverage against the number of labelled warm-up epochs from the new patient. Right: the corresponding mean prediction-set size. The horizontal line marks nominal coverage.

the deficit score) sits below the grid's best row and costs two thousandths of worst-patient coverage for APS and one point for the deficit score, relative to the best fixed value in hindsight; that is the price of choosing without the test cohort, and both are small against the eleven- and eight-point gains over pooled calibration.

One property of Equation (15) deserves an explicit check. The schedule

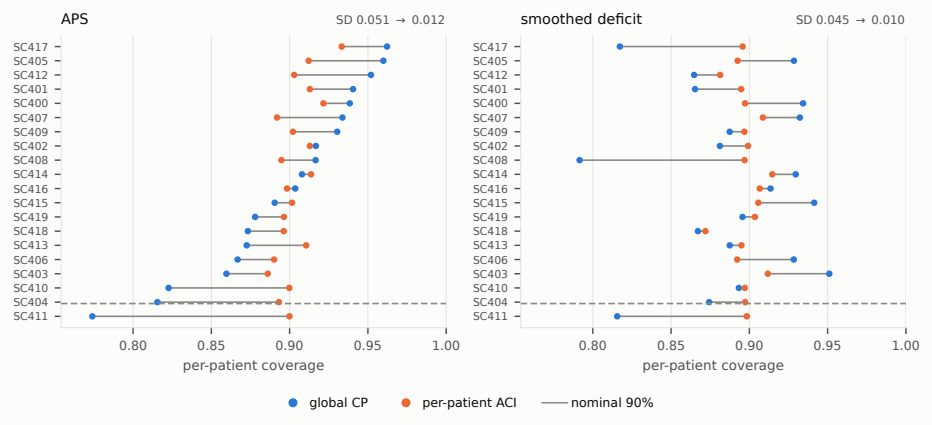


Figure 4: Per-patient coverage under pooled calibration and under per-patient ACI, one line segment per patient, sorted by pooled coverage. Patients to the left of the nominal line gain coverage; patients to the right give up coverage they did not need.

has $\sum_t \gamma_t^2 = \infty$ and therefore falls outside the square-summability condition under which the long-run ACI bound is proved, so the improvement above cannot be attributed to the theory. Repeating the identical experiment with $\gamma_t = \gamma_0/(t+2)^{0.6}$, which does meet the condition, leaves the conclusion in place and slightly weaker: worst-patient coverage 0.877 against 0.774 for pooled calibration, per-patient SD 0.015 against 0.051, at a mean set size of 2.51. The faster-decaying step adapts more slowly within a night, which is the expected price of a guarantee that the default schedule does not carry. That guarantee is Corollary 1, and it is asymptotic: it says that each patient's realised miscoverage converges to α as the stream grows, while a night contributes only a few hundred epochs, so it predicts the direction of the effect rather than its size. The default schedule is left in the paper because it is what an implementation would use; the poly-schedule run is there so that the reader can see the size of the effect under a schedule for which the corollary actually applies.

Figure 4 shows the mechanism directly. The reallocation is the point: the mean coverage is unchanged, and a fraction of patients end up a hair below nominal by construction, so the fraction-below statistic is not informative here — the dispersion and the worst patient are. A single stream over all patients cannot do this by construction: it returns one threshold for every patient, so it can move the marginal rate but not the per-patient dispersion, which is the reason we run one stream per patient. We therefore do not

Table 6: Per-patient adaptive conformal inference at $\alpha = 0.1$. Each patient is an independent stream; the step scale is chosen by leave-one-subject-out simulation on the calibration subjects and then applied unchanged to the test cohort.

Method	Cov.	Worst	SD	MAD_α	Below	Size	Sing.	Cons.
Global CP (APS)	0.898 ± 0.038	0.774	0.051	0.040	0.450	2.537 ± 0.207	0.167 ± 0.051	0.680 ± 0.037
Global CP (deficit)	0.889 ± 0.044	0.792	0.045	0.036	0.600	1.531 ± 0.325	0.841 ± 0.080	0.942 ± 0.018
Per-patient ACI (APS)	0.904 ± 0.010	0.886	0.012	0.009	0.500	2.491 ± 0.203	0.177 ± 0.060	0.675 ± 0.020
Per-patient ACI (deficit)	0.898 ± 0.007	0.872	0.010	0.007	0.700	1.575 ± 0.271	0.831 ± 0.072	0.940 ± 0.014

Coverage statistics merge per-subject coverage across folds and average each subject over the runs (folds × seeds × backbones). Mean LOSO-selected γ_0 over the runs: aps = 0.0623, deficit = 0.075. Set size and consistency are averaged over the same runs.

Table 7: Sensitivity of per-patient ACI to the step scale γ_0 at $\alpha = 0.1$. None of these rows is selected on test data; the deployable choice is the LOSO-selected one in Table 6.

Method	Coverage	Worst subj.	p_{10} subj.	Frac. below	Set size
Per-patient ACI (APS), $\gamma_0 = 0.005$	0.904 ± 0.026	0.810	0.864	0.400	2.566 ± 0.185
Per-patient ACI (APS), $\gamma_0 = 0.01$	0.908 ± 0.018	0.848	0.872	0.300	2.581 ± 0.197
Per-patient ACI (APS), $\gamma_0 = 0.02$	0.908 ± 0.011	0.875	0.887	0.400	2.553 ± 0.199
Per-patient ACI (APS), $\gamma_0 = 0.05$	0.902 ± 0.006	0.885	0.891	0.500	2.469 ± 0.177
Per-patient ACI (APS), $\gamma_0 = 0.1$	0.898 ± 0.004	0.888	0.890	0.750	2.409 ± 0.163
Per-patient ACI (deficit), $\gamma_0 = 0.005$	0.894 ± 0.030	0.828	0.859	0.600	1.579 ± 0.255
Per-patient ACI (deficit), $\gamma_0 = 0.01$	0.898 ± 0.021	0.859	0.862	0.600	1.613 ± 0.260
Per-patient ACI (deficit), $\gamma_0 = 0.02$	0.900 ± 0.014	0.860	0.884	0.600	1.616 ± 0.268
Per-patient ACI (deficit), $\gamma_0 = 0.05$	0.899 ± 0.007	0.873	0.887	0.550	1.590 ± 0.267
Per-patient ACI (deficit), $\gamma_0 = 0.1$	0.898 ± 0.004	0.882	0.889	0.650	1.562 ± 0.265

tabulate a single-stream variant; the comparison that matters is against the pooled threshold, which is exactly what a single stream converges to.

5.5. Sequence constructors: a negative result

Table 8 reports the dynamic-programming constructors alongside their per-epoch comparators. The first block uses every test epoch and includes the per-stage deficit threshold of Section 5.2, so that the score-family comparison is in one place; the constructor blocks use only the epochs inside complete blocks, and the per-epoch baseline immediately above the SeqCP-Margin block is recalibrated on exactly those epochs. Three observations follow, and they are the reason we do not present these constructors as a contribution.

At $\beta = 0$ the margin constructor is exactly per-epoch split CP on the deficit score (Proposition 2). The numerical check in the test suite confirms the identity to machine precision, and Table 8 confirms it at the level of realised coverage and set size: the $\beta = 0$ row and the baseline above it agree

to every reported digit. This is not a nuisance: it is what allows the $\beta > 0$ rows to be interpreted as the effect of the transition term alone.

At $\beta > 0$ the sets get larger. The block-level set is a union over admissible sequences, and the transition penalty makes low-transition sequences cheap; at $\beta \geq 0.5$ a constant sequence is admissible for every label, the projection is the whole label set, and sequence coverage is 1.000 by triviality. The simultaneous guarantee of Theorem 1 holds and is vacuous. Table 8's SeqCP-Block rows show this directly.

The transition structure is already in the score. The smoothed posterior that the deficit score is computed from has absorbed the same Markov structure that the dynamic program is meant to exploit. Re-imposing it at the level of the set adds looseness rather than information. This is why the efficiency gain in Section 5.2 comes from smoothing, not from sequence construction, and it is the substantive lesson of the ablation.

Figure 5 shows the consequence on a real recording: pooled APS produces sets that flicker at transitions, the deficit score produces stable singleton sets through stable stretches and wider sets at transitions, and per-patient ACI widens the sets for a patient whose pooled coverage was low.

5.6. Robustness to hardware shift and input corruption

Table 9 reports coverage under hardware shift: the model is trained on one Sleep-EDF cohort (cassette or telemetry), calibrated on four held-out subjects of that cohort, and tested on the other. The ranking of scores changes. The deficit score, the most efficient in-cohort, is also the most fragile: pooled coverage falls from 0.889 to 0.835 (cassette to telemetry) and 0.878 (telemetry to cassette), and the worst patient falls to 0.714 and 0.678. Two mechanisms contribute — the forward–backward pass uses transition matrices estimated from source-cohort labels, and the posterior sharpness changes with the signal statistics of the target cohort. Pooled APS holds coverage better — it over-covers at 0.933 in one direction and sits at 0.886 in the other — but its worst patient is at 0.799 and 0.789, and it pays with the largest sets (2.68 and 2.20 labels per epoch). Per-patient ACI, which adapts on the target stream and therefore does not assume the calibration cohort matches the target population, brings marginal coverage back to between 0.896 and 0.913 in both directions and lifts the worst patient to 0.883 and 0.872 for APS and to 0.878 in both directions for the deficit score; for the deficit score this costs 32% and 6% in mean set size ($1.40 \to 1.84$ and $1.77 \to 1.87$).

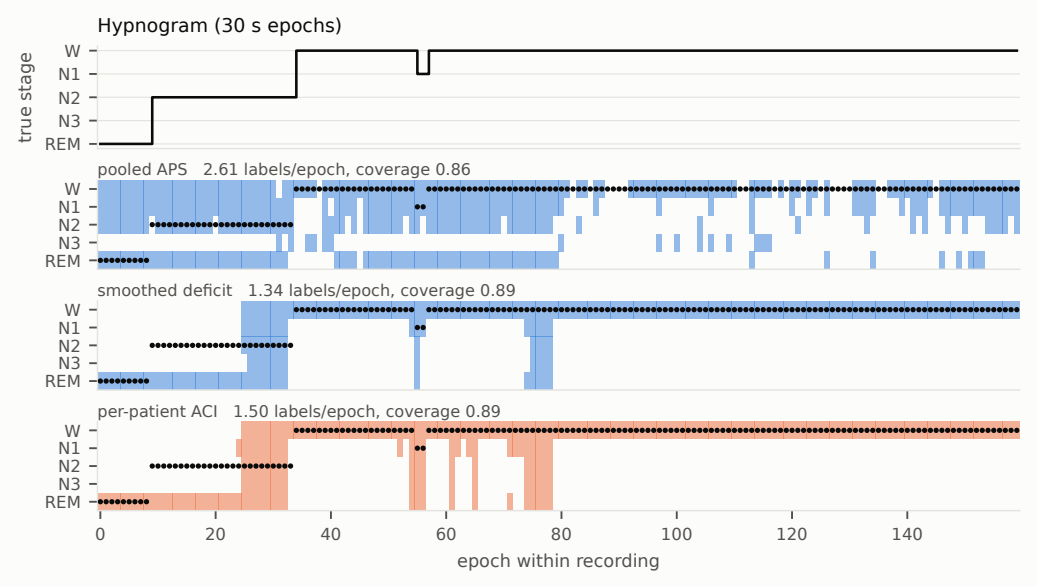


Figure 5: A 160-epoch window of one test recording (subject SC409), chosen because it spans a stage transition. Top: the scored hypnogram. Panels: one row per stage, with a shaded cell where that stage is in the prediction set under pooled APS, smoothed deficit, and per-patient ACI. The black dots repeat the true stage inside each panel, so an epoch is covered exactly when its dot lies on a shaded cell; the label count and coverage printed with each panel are over this window only.

Table 10 applies controlled corruptions to the test signals while calibration stays clean, and contrasts three thresholds: the clean one, one refitted on identically corrupted calibration signals, and one adapted online by per-patient ACI. The loss is not monotone in severity across corruption families, and it is not always a loss: a circular time shift, which preserves the spectrum, moves coverage by less than a thousandth and leaves accuracy unchanged, and zeroing a fifth of the epochs costs 1.3 points of coverage for twelve points of accuracy. Where the corruption bites, the clean threshold fails catastrophically (0.497 coverage at 0 dB SNR, 0.141 under 2σ baseline drift), and both adaptations restore coverage (0.892 and 0.929 by recalibration; 0.893 and 0.890 by ACI). Recalibration restores more than the number: corrupted calibration and corrupted test signals are exchangeable again, so the split-conformal guarantee applies to the corrupted data. ACI makes no such argument — it carries the long-run property of Section 3.9 — and its values here are measured, not proved.

Table 8: Temporal conformal prediction at $\alpha = 0.1$, block length $L = 10$. The first four rows use every test epoch; the rows below them use only the epochs inside complete blocks, and the per-epoch baseline above the SeqCP-Margin block is recalibrated on exactly those epochs.

Method	Coverage	Set size	Singleton	Review	Consistency	Seq. coverage
Smoothed deficit (all epochs)	0.889 ± 0.044	1.531 ± 0.325	0.841 ± 0.080	0.159 ± 0.080	0.942 ± 0.018	–
Smoothed deficit, per stage	0.903 ± 0.034	1.741 ± 0.252	0.506 ± 0.104	0.494 ± 0.104	0.884 ± 0.016	–
Smoothed APS (all epochs)	0.903 ± 0.037	2.223 ± 0.170	0.229 ± 0.053	0.712 ± 0.062	0.753 ± 0.036	–
Per-epoch APS (all epochs)	0.898 ± 0.038	2.537 ± 0.207	0.167 ± 0.051	0.765 ± 0.060	0.680 ± 0.037	–
Smoothed deficit, $L = 10$ epochs	0.889 ± 0.045	1.535 ± 0.326	0.840 ± 0.080	0.160 ± 0.080	0.942 ± 0.018	–
SeqCP-Margin L10, $\beta = 0$	0.889 ± 0.045	1.535 ± 0.326	0.840 ± 0.080	0.160 ± 0.080	0.942 ± 0.018	0.668 ± 0.089
SeqCP-Margin L10, $\beta = 0.5$	0.890 ± 0.045	1.941 ± 0.616	0.688 ± 0.215	0.312 ± 0.215	0.865 ± 0.061	0.682 ± 0.096
SeqCP-Margin L10, $\beta = 1$	0.891 ± 0.045	1.993 ± 0.634	0.655 ± 0.230	0.345 ± 0.230	0.864 ± 0.060	0.688 ± 0.096
SeqCP-Margin L10, $\beta = 2$	0.889 ± 0.045	2.016 ± 0.625	0.666 ± 0.166	0.334 ± 0.166	0.863 ± 0.062	0.680 ± 0.080
SeqCP-Block L10, $\beta = 0$	0.919 ± 0.032	4.570 ± 0.167	0.005 ± 0.002	0.914 ± 0.033	0.909 ± 0.035	0.917 ± 0.033
SeqCP-Block L10, $\beta \geq 0.5$	1.000 ± 0.000	5.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000

"Review" is the fraction of epochs whose set contains more than one stage and would need adjudication; "Consistency" is the mean Jaccard overlap of adjacent prediction sets within a recording; "Seq. coverage" is the fraction of blocks whose every epoch is covered. The SeqCP-Block rows at $\beta = 0.5, 1, 2$ coincide exactly; $L = 20$ and $L = 30$ change no conclusion; the full sweep is in the released JSON. SeqCP-Margin at $\beta = 0$ coincides with the per-epoch baseline of the same name to every reported digit, which is the numerical check of Proposition 2; realised coverage is an expectation over calibration draws, not a bound for a single calibration set.

The accuracy column is the reason to read that recovery carefully. Under the same two corruptions the classifier itself is at 0.257 and 0.252, against 0.802 clean. What the recalibrated threshold restores is the truthfulness of the set, not the model, and it pays for the truth by reporting larger sets (3.44 and 3.93 labels per epoch against 3.00 clean). ACI, which never sees corrupted calibration data, reaches the same coverage at 3.30 and 3.32. A deployment that reports only coverage would call this a recovery; one that reports coverage and set size sees a model that has stopped working and a predictor that says so — which is the behaviour we want, provided the set size is reported.

5.7. Extended cohort

The 78-subject cohort increases the number of test subjects per fold from four to fifteen or sixteen (and the calibration cohort from four to sixteen), and lets the worst-patient statistics be estimated with less noise. Table 11 reports the same headline comparison; the ordering of methods is unchanged and the pooled-to-per-patient gap is larger, from 0.641 to 0.879 (APS) against

Table 9: Coverage under hardware shift. The in-cohort rows are the cassette cohort over all runs of the main matrix; the cross rows train on one cohort, calibrate on four held-out subjects from it, and test on the other cohort, at seed 0 for both backbones, with the statistics merged over the five folds and two backbones. The ± is the standard deviation over the runs contributing to the row (thirty in-cohort, ten cross-cohort), so it measures run-to-run variation, not uncertainty over patients. Per-patient ACI uses the leave-one-subject-out step size selected within the source cohort.

Scenario	Method	Coverage	Worst patient	p_{10}	Set size
In-cohort	APS, pooled	0.898 ± 0.038	0.774	0.822	2.537 ± 0.207
	APS, per-patient ACI	0.904 ± 0.010	0.886	0.892	2.491 ± 0.203
	Deficit, pooled	0.889 ± 0.044	0.792	0.817	1.531 ± 0.325
	Deficit, per-patient ACI	0.898 ± 0.007	0.872	0.891	1.575 ± 0.271
Cassette → telemetry	APS, pooled	0.933 ± 0.028	0.799	0.869	2.684 ± 0.066
	APS, per-patient ACI	0.913 ± 0.010	0.883	0.894	2.387 ± 0.018
	Deficit, pooled	0.835 ± 0.026	0.714	0.747	1.397 ± 0.083
	Deficit, per-patient ACI	0.896 ± 0.001	0.878	0.888	1.843 ± 0.099
Telemetry → cassette	APS, pooled	0.886 ± 0.003	0.789	0.838	2.198 ± 0.079
	APS, per-patient ACI	0.903 ± 0.001	0.872	0.883	2.292 ± 0.152
	Deficit, pooled	0.878 ± 0.001	0.678	0.820	1.772 ± 0.280
	Deficit, per-patient ACI	0.898 ± 0.005	0.878	0.880	1.874 ± 0.264

Table 10: Coverage at the most severe level of each input corruption, applied to the test signals only. Accuracy is the corrupted model's own epoch accuracy and does not depend on any threshold. "Clean threshold" is the split-conformal threshold fitted on clean calibration data; "recalibrated" refits the threshold on calibration signals corrupted identically, and the following column is the mean set size it returns; "per-patient ACI" adapts the threshold online on the corrupted stream using the clean calibration set, again with its mean set size. One representative run (EEGNet, fold 0) is reported because the corruption is a property of the model and the split rather than of the fold.

Corruption	Severity	Accuracy	Clean threshold	Recalibrated	Recal. size	Per-patient ACI	ACI size
None	–	0.802	0.936	0.936	3.00	0.936	3.00
Noise	0 dB SNR	0.257	0.497	0.892	3.44	0.893	3.30
Amplitude scale	×1.5	0.717	0.831	0.964	2.90	0.896	2.38
Baseline drift	2σ wander	0.252	0.141	0.929	3.93	0.890	3.32
Circular time shift	5 s	0.802	0.936	0.937	3.00	0.902	2.65
Epoch dropout	20%	0.683	0.923	0.921	3.17	0.902	3.06

0.774 to 0.886 in the 20-subject cohort. Part of that widening is mechanical, since the minimum is taken over nearly four times as many patients, but the per-patient SD says the same thing without an extremum: 0.067 against 0.016 for APS and 0.053 against 0.013 for the deficit score, both larger than the corresponding 0.051 and 0.045 against 0.012 and 0.010 of Table 6.

Table 11: The headline comparison of Table 2 repeated on the 78-subject cohort, at seed 0. Test blocks of fifteen or sixteen subjects make each fold's estimate less sensitive to a single night, while the minimum over all subjects is naturally more extreme than in the smaller cohort.

Method	Coverage	Worst subj.	p_{10} subj.	SD	Below	Set size	Singleton
Random-epoch split	0.900	–	–	–	–	2.898	–
Pooled (APS)	0.898 ± 0.036	0.641	0.822	0.0667	0.33	2.890 ± 0.231	0.059 ± 0.031
Pooled (RAPS)	0.905 ± 0.031	0.751	0.854	0.0380	0.37	1.845 ± 0.271	0.306 ± 0.204
Mondrian (per class)	0.908 ± 0.030	0.681	0.840	0.0599	0.33	2.306 ± 0.111	0.261 ± 0.030
kNN subject (APS)	0.893 ± 0.032	0.678	0.843	0.0627	0.42	2.776 ± 0.170	0.072 ± 0.029
Warm-up SACP, $m = 80$	0.915 ± 0.043	0.631	0.840	0.0704	0.24	3.075 ± 0.414	0.073 ± 0.031
Warm-up SACP, $m = 320$	0.898 ± 0.029	0.700	0.852	0.0481	0.38	2.847 ± 0.308	0.078 ± 0.031
Oracle per patient	0.906 ± 0.011	0.900	0.901	0.0137	0.00	2.758 ± 0.181	0.094 ± 0.025
Per-patient ACI (APS)	0.917 ± 0.007	0.879	0.898	0.0163	0.14	2.898 ± 0.152	0.072 ± 0.028
Per-patient ACI (deficit)	0.897 ± 0.004	0.861	0.884	0.0133	0.54	1.657 ± 0.157	0.805 ± 0.045

"Coverage" is marginal over epochs (mean ± SD over the runs); "Worst subj." is the minimum per-subject coverage over the runs; "Below" is the fraction of subjects below $1 - \alpha$; "SD" is the standard deviation of per-subject coverage. Each subject contributes one value per run, the runs being five folds × one seed × two backbones; set size and singleton rate are averaged over the same runs.

6. Discussion

6.1. Why smoothing helps and the dynamic program does not

The two temporal results are not in tension; they follow from the same fact. Split CP calibrates a *scalar* per epoch and produces a set by thresholding it. Any transformation of the model output that makes the true label's scalar smaller relative to the other labels — which the forward–backward posterior does, because it integrates evidence over the whole recording — improves efficiency without touching validity. The dynamic program does something different: it minimises over *paths*, and a path-level minimum is always at most a point-level one, so the resulting level comparison is looser rather than tighter. The transition structure that the DP is meant to exploit has already been absorbed into the smooth posterior at the point where the score is computed. This is why the ablation at $\beta = 0$ is the decisive one: it shows the two constructions are identical there, and positive β only adds looseness.

The practical consequence is a simpler recipe than the one we set out to build: smooth, score the deficit, calibrate on subjects, adapt per patient. That recipe needs no dynamic program and no block-level calibration. It also separates two sources of ambiguity. Irreducible disagreement over an

N1 or transition epoch remains visible as a larger set, whereas a mismatch between a new patient and the calibration cohort is corrected by the adaptive stream. This is the operational counterpart of the aleatoric–epistemic distinction discussed for sleep staging [27].

6.2. What per-patient adaptation costs and buys

Per-patient ACI requires labels on a delay. In sleep staging the delay is the technologist's review, so the feedback assumption is realistic, but it does mean that the guarantee is about the *realised* sequence of predictions and not about a prediction made before any review has happened. The first epochs of a new patient are therefore predicted with the pooled threshold, and adaptation only begins to pay after a few dozen epochs. The coverage vectors in Section 5.4 show the mechanism: easy patients lose coverage that they did not need, hard patients gain coverage that they did, and the aggregate changes by less than a point. This reallocation is the point, not a side effect.

The step scale is the one hyperparameter, and it is chosen without test labels by simulating held-out streams inside the calibration cohort. That choice is conservative: it optimises the mean one-sided shortfall rather than the worst patient, so a deployment that cares about the single worst patient should select on that criterion instead; the released code exposes the choice as `-criterion worst`.

6.3. Reading the sets in a review workflow

A single coverage number is not enough to evaluate a set-valued stager, for three reasons that this study makes concrete. It is marginal over epochs and therefore blind to the patient whose sets are systematically too small. It is satisfied trivially by large sets, so it must be read against set size and review workload. And it says nothing about temporal behaviour, which in a hypnogram is the difference between a usable display and a flickering one. We would therefore argue that a conformal sleep-staging paper should report per-patient coverage, the fraction of patients below nominal, set size, and a temporal consistency statistic, and should use subject-wise folds that partition the cohort.

Those four numbers translate directly into the review step. A night of eight hours contains 960 epochs; at the 0.76 workload of pooled APS, 730 of them would be sent back for adjudication, whereas the smoothed deficit score sends about 150, plus the empty sets that also return to the reviewer. The size of a non-singleton set is itself informative, because it separates "N2

or N3" from "any of four stages": only the first is a question a technologist can settle by re-reading a page of the record. It is worth being explicit that a set of two stages is not automatically a weaker statement than a point label. Inter-scorer studies show that two trained technologists frequently disagree on N1 and on transition epochs [1], so an admitted ambiguity of that size is commensurate with the reference standard against which any stager is judged, whereas a confident point prediction that is wrong is not.

The resulting deployment pattern is short. The stager is frozen; its HMM transitions come from the training labels; one pooled threshold is calibrated on the handful of scored nights a unit already has; and the per-patient adaptive stream runs during review, consuming each stage label after it has been confirmed. Nothing in the procedure is fitted to the patient being scored, which is the property that makes it usable in a unit that cannot collect labelled epochs in advance.

6.4. Limitations

Single channel, five classes. All experiments use Fpz–Cz only and the five AASM stages. Multi-channel recordings, and problems that use the finer R&K stages, will have different score distributions; the methods apply unchanged, but the specific numbers should not be extrapolated.

Cohort size. The main protocol tests four subjects per fold, and the merged estimate is over 20 subjects. This is small, and the seed spread we report is a spread over training runs rather than over patient cohorts. The extended cohort in Section 5.7 is a partial answer, but a multi-centre study with several hundred patients would be needed to put a confidence interval on the worst-patient statistic.

Feedback assumption. ACI needs the label after the prediction. In a fully automatic deployment with no review, the stream is unavailable and only the pooled or warm-up rules apply.

N1. The backbones used here reproduce the known weakness of the literature: N1 F1 is low. The consequence for the sets is asymmetric. Pooled APS over-covers N1 because its threshold is loose, whereas the smoothed deficit score omits N1 in more than half of the N1 epochs (coverage 0.43 against nominal 0.90), because the model is confidently wrong on exactly those epochs. This qualifies the efficiency claim of Section 5.2; thresholding

the same score per stage restores every stage to nominal coverage, and Table 3 reports what that costs in set size (1.74 against 1.53 labels). A stronger backbone would shift every number; we expect, but have not shown, that it would leave the ordering of the methods unchanged, because the ordering is driven by the protocol rather than by the model.

No clinical evaluation. We make no claim that a technologist working with these sets is faster or more accurate than one working with point predictions. Review workload is a proxy, and it should be validated in a reader study before deployment.

7. Conclusions

Conformal prediction is a natural way to give a clinician a statement about a sleep-stage prediction that is honest under subject shift. We have shown that the statement the standard method makes is the wrong one — pooled over epochs rather than per patient — and that two simple changes together address it. Scoring the smoothed posterior deficit shrinks the average set by about forty per cent at comparable coverage and cuts the fraction of epochs needing manual review from 0.76 to 0.16 (to 0.49 if the threshold is conditioned on the stage, which holds every stage at nominal), and running an independent adaptive stream per patient moves coverage from easy patients to hard ones without using any test labels for tuning. The dynamic-programming sequence constructors that motivated this work turn out not to help, for a reason we can prove, and we report that result explicitly, together with the exact protocol and code needed to check it.

The practical form of the result is a stager that reports, epoch by epoch, the stages it cannot rule out, and a number of epochs it cannot settle. That does not replace the technologist's review; it tells the technologist where the review is worth spending.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics statement

This study used publicly available, de-identified Sleep-EDF recordings from PhysioNet. No new human-subject data were collected, and additional ethics approval was therefore not required.

Data availability

Sleep-EDF and the Sleep-EDF telemetry cohort are publicly available from PhysioNet. Code reproducing every table and figure, together with the exact command lines and released result summaries, is available at `https://github.com/YuhaoPeng1103/SleepCP`.

CRediT authorship contribution statement

Zhiwen Song: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft. **Yuhao Peng:** Supervision, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, manuscript organisation, and consistency checks. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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